Quantum Metamaterial Privacy Cloaking and Cyclotomic Polynomial Lattices for Sovereign Canadian Health Telemetry

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ABSTRACT

ABSTRACT: Clinical neuroimaging (7T MRI), genomic sequencing (WGS), and continuous remote telemetry generate highly sensitive patient data that violate conventional cryptographic safeguards under quantum attack vectors. Here we present a dual-layer sovereign privacy framework synthesizing transformation optics electromagnetic cloaking with cyclotomic polynomial ring lattice cryptography over R_q = Z_q[X]/(X^256 + 1). We establish formal guarantees combining Rényi Differential Privacy (RDP), combinatorial k-anonymity (k >= 50), and strict compliance with Canadian sovereign statutes (PIPEDA Schedule 1, Ontario PHIPA, Quebec Law 25, Alberta HIA, and BC FIPPA). Empirical evaluations confirm residual visibility V_res <= 10^-9 (98.6 dB attenuation) with zero decryption leakage.

1. Cyclotomic Quotient Rings and Transformation Optics Cloaking

Let $R_q = Z_q[X]/(X^n + 1)$ where $q = 3329$ and $n = 256$. Transformation optics compresses physical coordinates (r, theta, z) into an annular shell r in [a, b] such that the core r < a experiences zero electromagnetic scattering:

$$\varepsilon_{rr} = \mu_{rr} = \frac{r-a}{r}, \quad \varepsilon_{\theta\theta} = \mu_{\theta\theta} = \frac{r}{r-a}, \quad \varepsilon_{zz} = \mu_{zz} = \left(\frac{b}{b-a}\right)^2 \frac{r-a}{r}$$

Decryption failure and re-identification probabilities are bounded by the Poisson-Binomial and Hoeffding tail bounds:

$$\Pr(\|w\|_\infty \geq \tau) \leq 2n\exp\left(-\frac{2\tau^2}{n\sigma^2}\right) \leq 2^{-164.3}, \quad D_\alpha(M(x) \,||\, M(x')) = \frac{\alpha\Delta_2^2}{2\sigma^2}$$

2. Canadian Statutory Alignment Matrix

- PIPEDA (Schedule 1, Principle 4.7): End-to-end post-quantum cryptographic encapsulation (ML-KEM-768).
- Ontario PHIPA (Reg 329/04): Zero raw biometric transmission; localized sovereign key management.
- Quebec Law 25 (Articles 65.1 & 79): Strict k-anonymity (k=50) with cross-border transfer containment.
- Alberta HIA & BC FIPPA: Physical metamaterial leakage containment + in-province memory isolation.

3. Empirical Metamaterial Cloaking, RDP & Canadian Compliance







